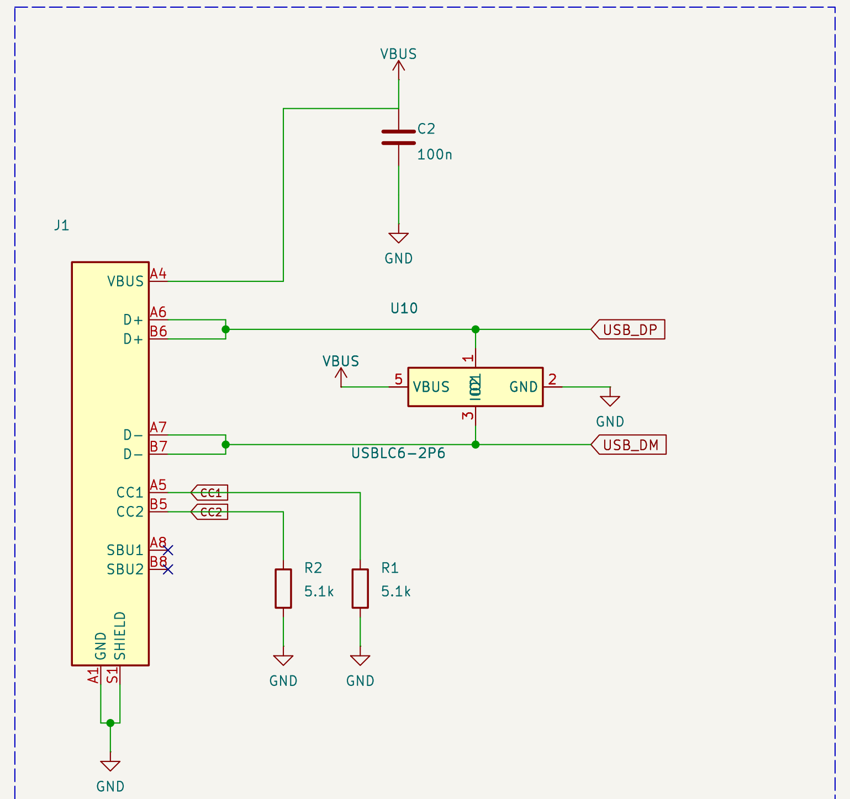
bluebuzz — generated schematic

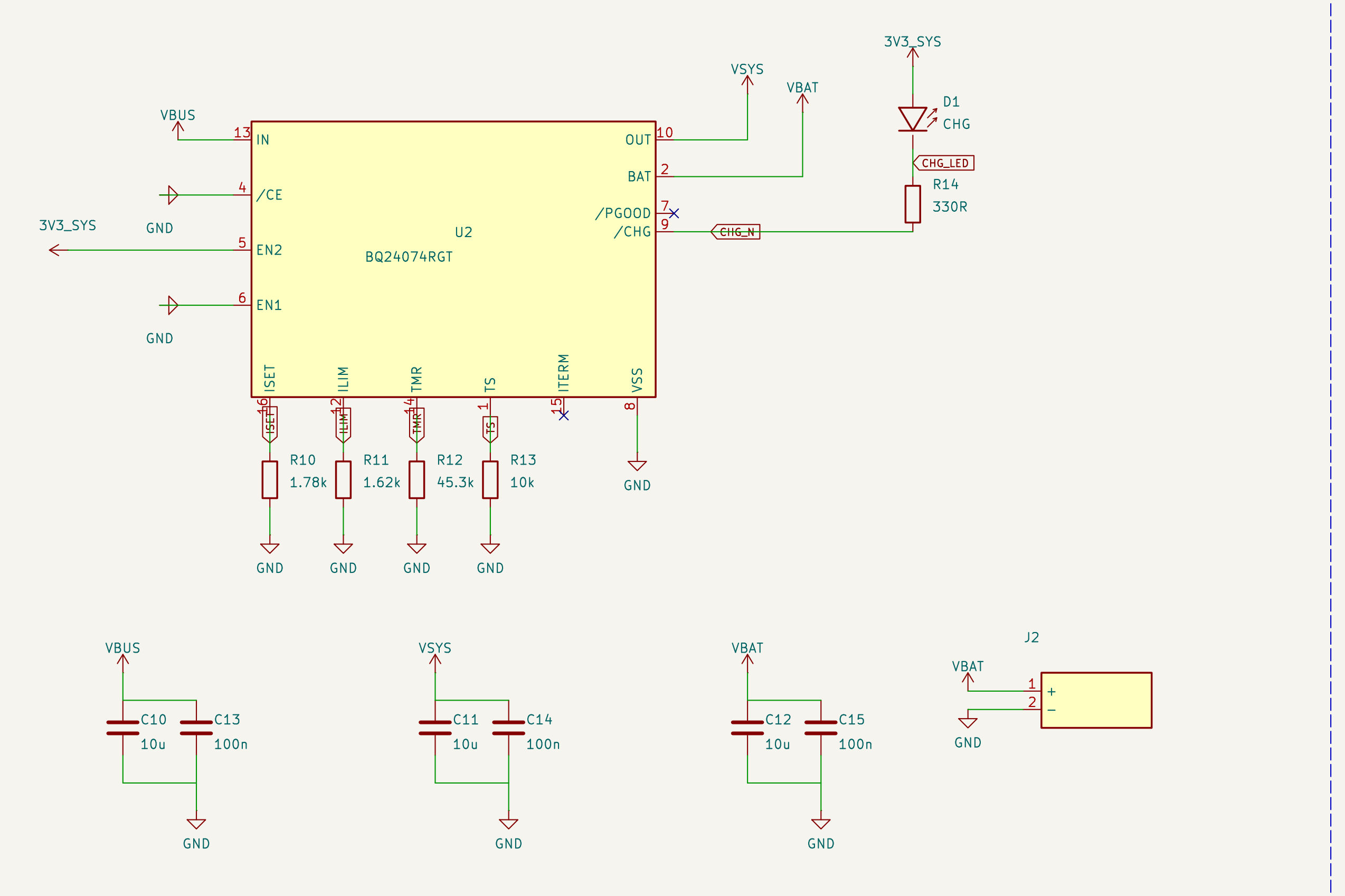
66 components / 42 nets / 4-layer. Bluetooth vibrator controller: nRF52840 module, USB-C charger with power path, two low-side motor channels from VSYS, HX711 load-cell ADC and a pressure-sensor port on a switched 3.3 V rail. Power rails are power symbols; only cross-block signals use global labels. Generated from scripts/circuit.py — do not hand-edit; rebuild with `python3 build.py --only schematic`.

Block 1 — USB-C input and ESD



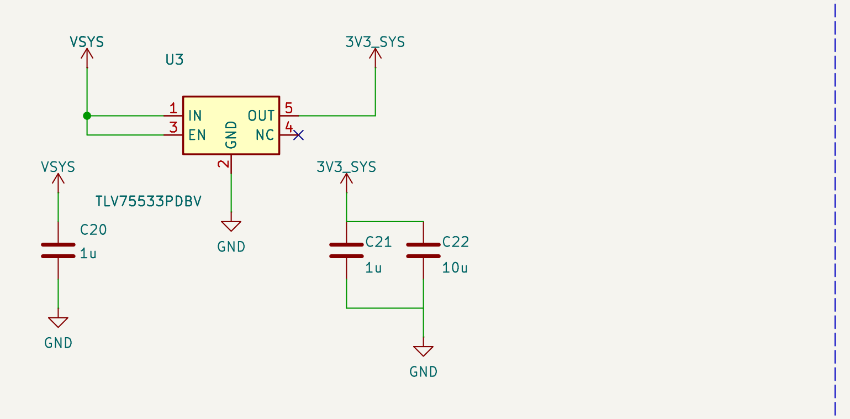
A6/B6 and A7/B7 are the same signal on a USB-C receptacle and must be tied, or the board only enumerates in one cable orientation. 5.1k on EACH CC line advertises a default-current sink. The USBLC6-2P6 (SOT-666) clamps VBUS and protects D+/D-, not CC — accepted, CC is short-traced.

Block 2 — BQ24074 charger with power path



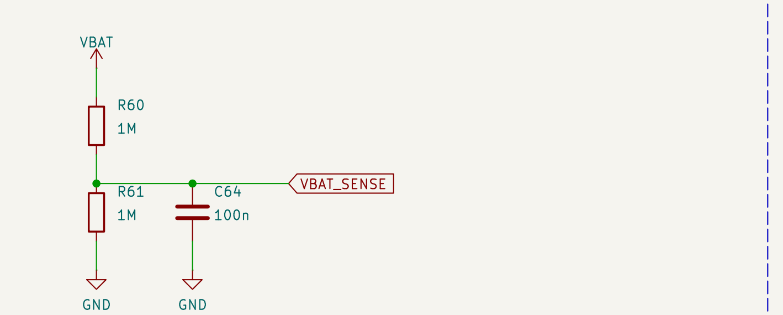
EN2 = 1, EN1 = 0 selects the R_ILIM input limit (1610/1.62k = 0.99 A). R12 = 45.3k gives a 6.0 h safety timer (10 x 48 s/kohm); the legal window is 18–72k. ITERM is left open on purpose: pin 15 is ITERM on the '74 and open selects the 10% default. VSYS is the power-path output and feeds the LDO AND the motor, so the toy runs on the cable with a flat cell. The PowerPad carries up to 1.0 W and must go to a stitched pour. J2 is the cell.

Block 3 — system LDO, always on



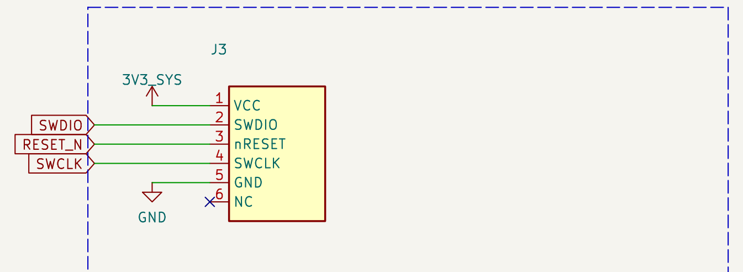
SOT-23-5: 1 = IN, 5 = OUT, EN tied to IN. 25 uA Iq. C22 is bulk for the radio's TX bursts. The motor does NOT run from this rail (Block 4 takes VSYS directly).

Battery voltage sense



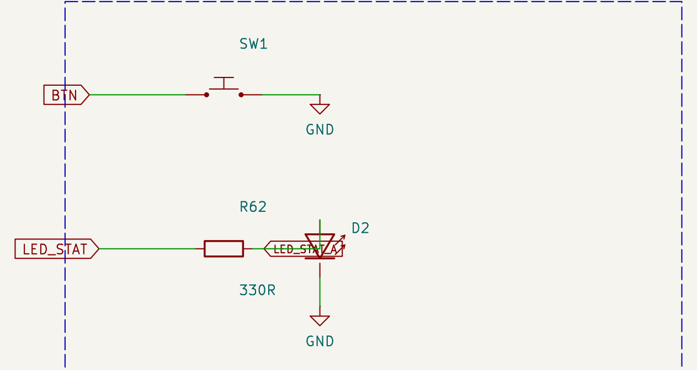
1M/1M draws 2.1 uA and needs no switch. C64 sources the SAADC sampling charge from the 500k source impedance — use a long acquisition time. AIN0 = P0.02, module pad 7.

SWD — Tag-Connect TC2030-IDC-NL (back side)



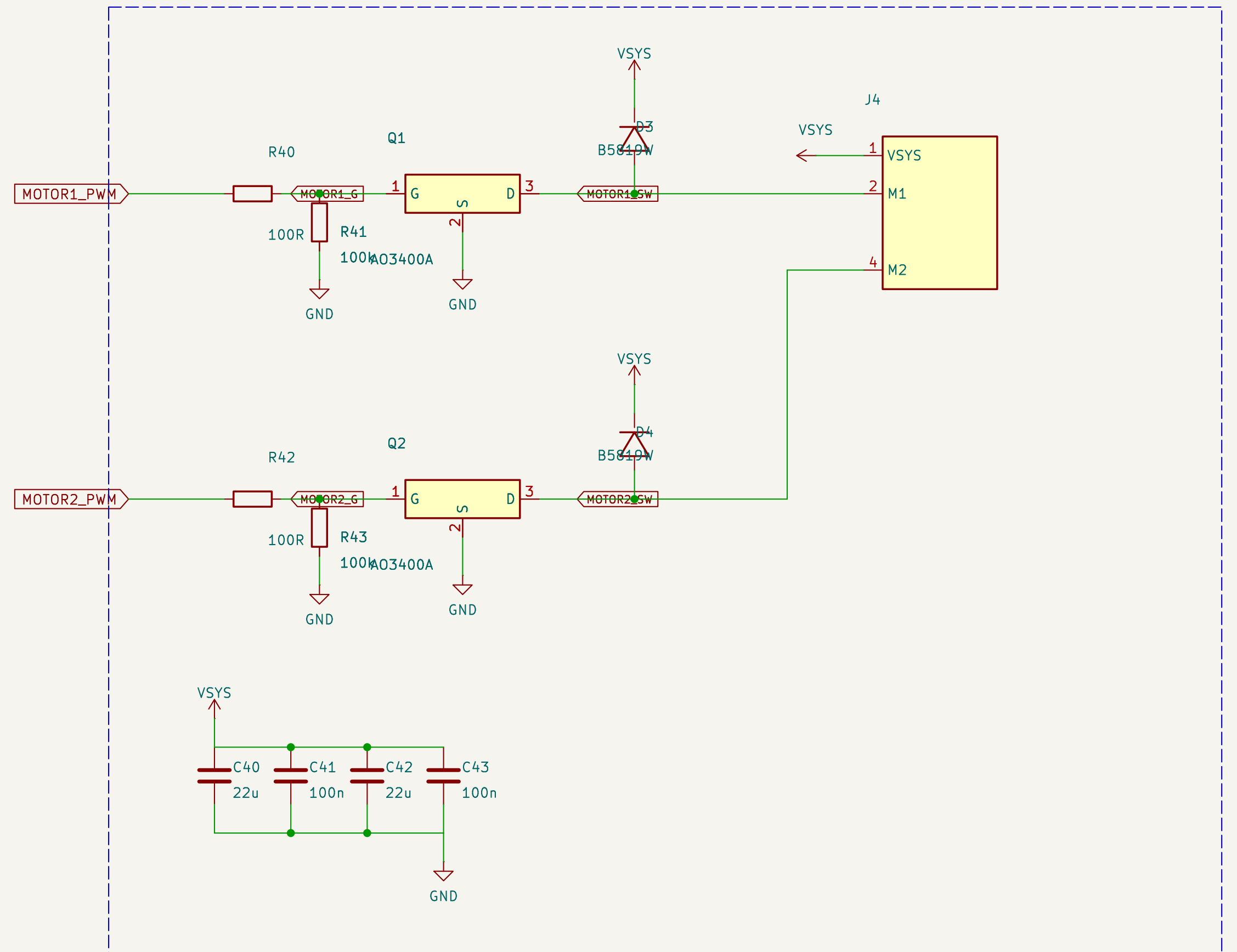
Pads only, on the BACK of the board where the plug's 9.9 x 7.2 mm head has room. Standard ARM order.

User button and status LED



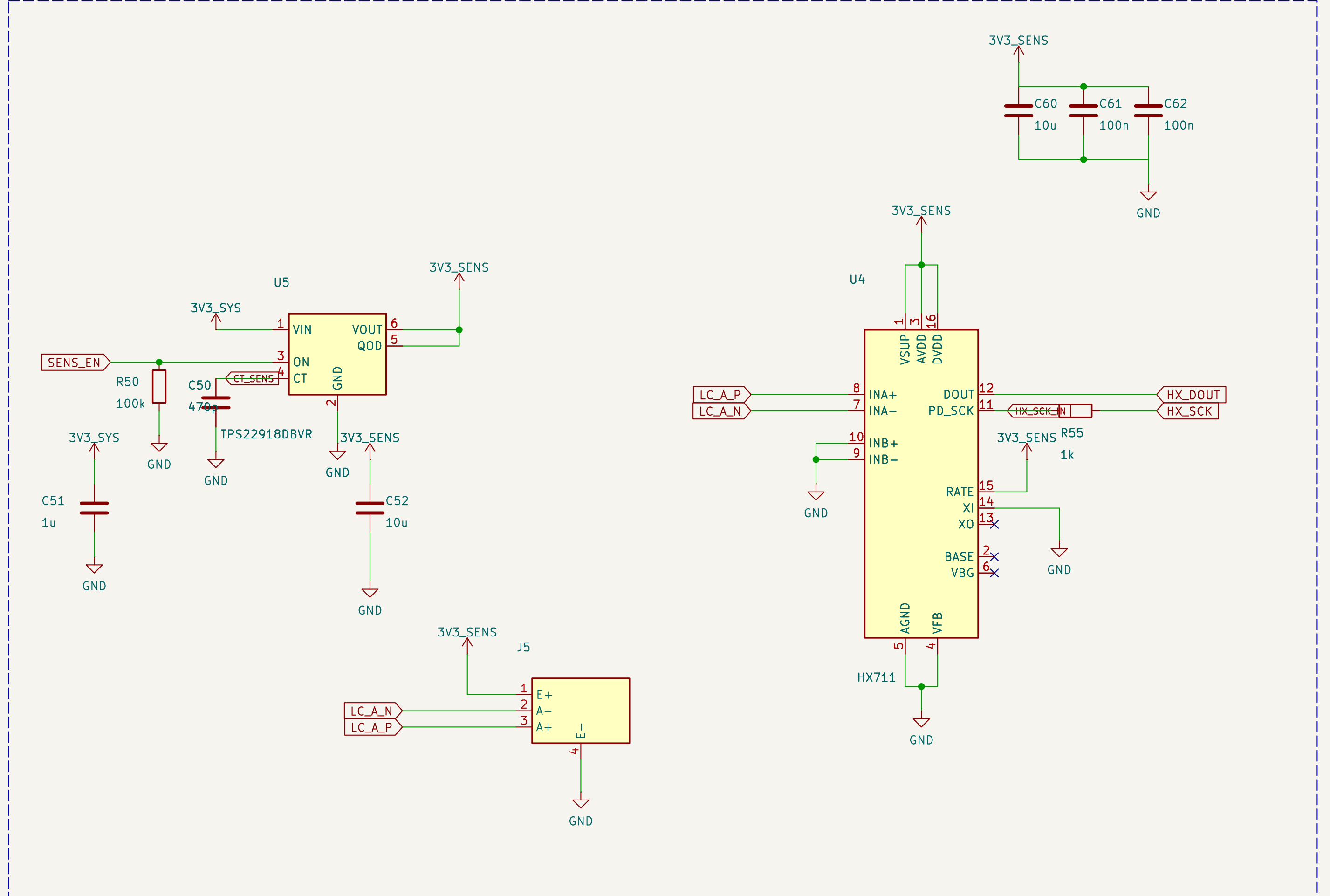
Side-actuated button on the left board edge, internal pull-up on P0.13. Green 0603 at ~1.5 mA from 330R; drive P0.24 high-drive.

Block 4 — motors: two low-side FET channels from VSYS



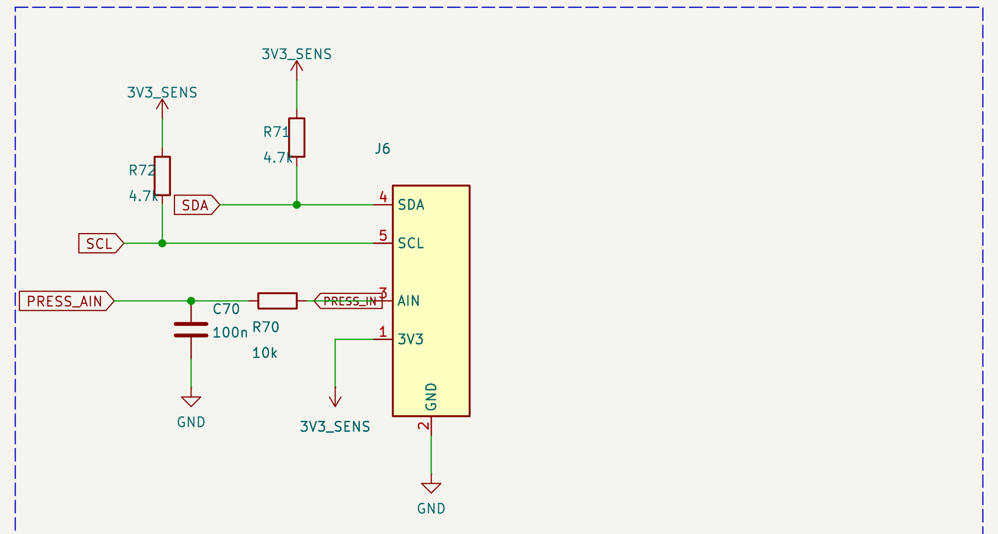
Two identical PWM channels: AO3400A low side (30 mOhm from a 3.3 V gate), B5819W 1 A Schottky across each motor for the off-time current, 2 x 22 u of bulk at the connector so the PWM ripple stays off the LDO's input. R41/R43 matter: an nRF GPIO floats from reset until firmware drives it, and a floating gate is a toy that buzzes on the bench. J4 is one 4-way JST PH for both leads: 1 VSYS, 2 M1, 3 VSYS, 4 M2. Each channel is sized for 1 A.

Blocks 5 + 6 — switched sensor rail, HX711 load-cell ADC and port



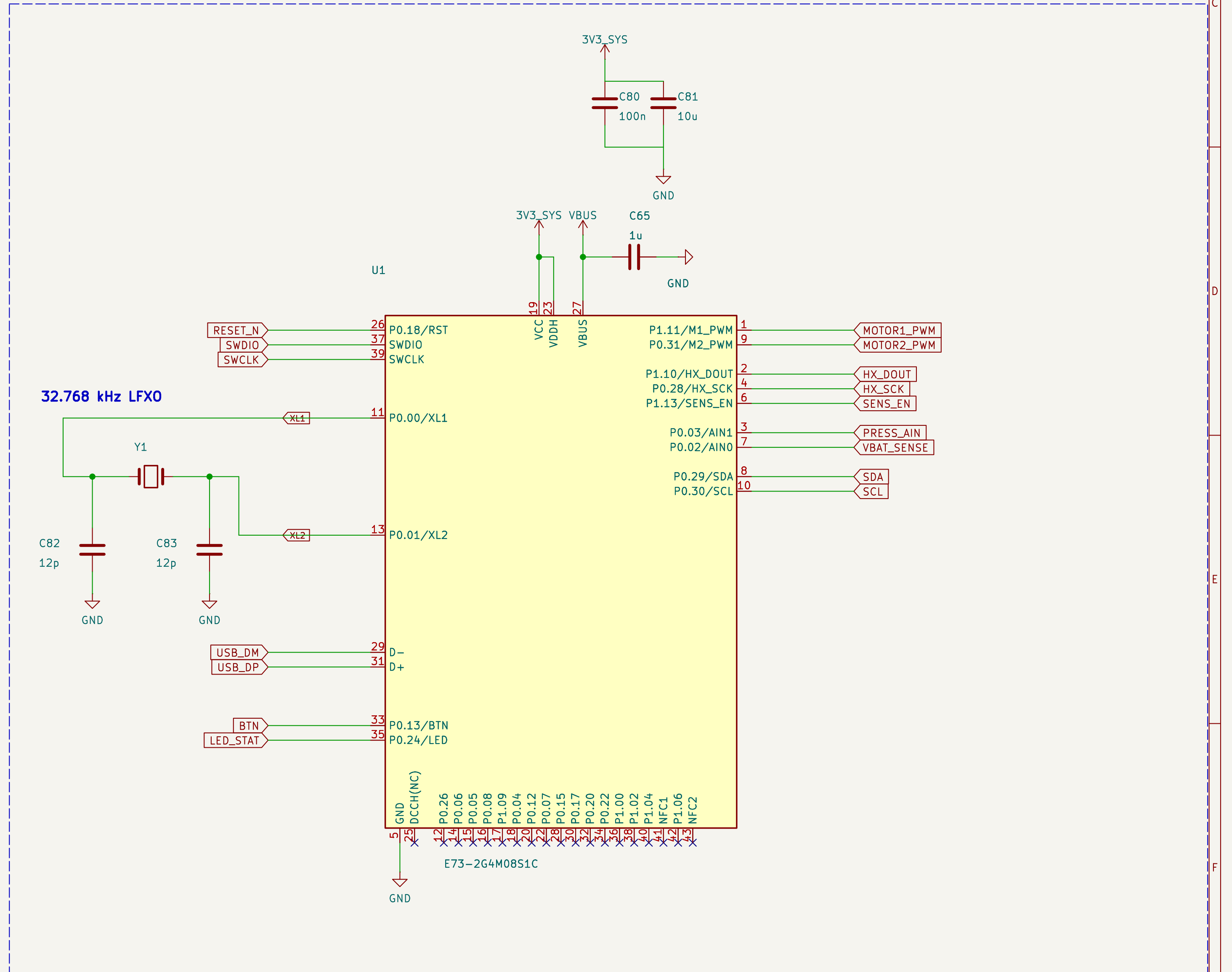
The HX711 runs WITHOUT its analog regulator: VSUP tied to AVDD, VFB to ground, BASE and VBG open (datasheet, Power Supply Options). XI = 0 selects the on-chip oscillator; RATE = DVDD is 80 SPS. Full scale is +-0.5 AVDD/128 = +-12.9 mV; a 2 mV/V cell on 3.3 V excitation gives 6.6 mV. The whole rail is switched (U5) because a 350 ohm bridge draws 9.4 mA forever; DVDD is therefore switched too, so firmware must tristate HX_SCK before gating, and R55 caps the current into an unpowered pin if it does not. QOD to VOUT or the rail never discharges; 100k so ON cannot float. J5: E+ A- A+ E-.

Block 7 — pressure sensor port (analog or I2C)



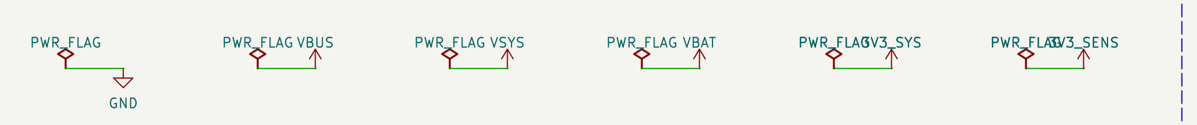
One 5-way port, two kinds of sensor. Analog (a ratiometric 0..3.3 V part): its output goes through 10k/100n (~160 Hz) into AIN1 = P0.03; the 10k also limits what a 5 V sensor plugged in by mistake can push into the pin's clamps. I2C (a digital barometric or gauge sensor): 4.7k pull-ups on 3V3_SENS, so an unpowered sensor is not back-fed through them. The nRF52840 puts TWIM on any pins.

Block 8 — nRF52840 module (E73-2G4M08S1C, onboard antenna)



43 pads: 28 castellated plus 15 bottom-only interior pads, so this board needs a stencil. Every GPIO used here is on an EDGE pad; only GND (24) is interior. The ceramic antenna is at the pad-free end of the body: that end sits at the board edge with no copper under it on any layer. Pad 26 is P0.18, the only pin the nRF52840 can use as nRESET. Y1 is Nordic's reference LFXO: a 3215 9 pF-CL part with 12 pF NP0 loads.

Power flags — the ERC root for each rail



These carry no copper and appear in no netlist. They exist because every rail here is fed through a pin the symbols declare passive, so ERC would otherwise report six undriven power nets.

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